Formalizing Resistance in Social Reward Economy

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October 12, 2026

1 Introduction

In this report, I give a formal definition of resistance within a social reward economy. *Learning Norms* gives a formal definition of a single reward economy and describes a model that converges reliably to the welfare optimal norm. The paper *Social Reward Economies* gives a description of how these ideas can be extended to cases where more than one norm are in competition and multiple social reward economies exist in the same system. This report aims to formalize a resistant subgroup as defined in *Social Reward Economies* with a similar structure to the model in *Learnin Norms* under certain simplifying assumptions.

2 Set up

The basic setup for the system is as follows:

- There is a group of agents G.
- A set of states S where $s \in S$ has probability $p(s)$ of occuring.
- For each $s \in S$ there is a set $A(s)$ of possible actions in state s.
- A policy is function that maps each $s \in S$ to a probability distribution over $A(s)$. Fix a dominant policy π_d.
- Each agent $i \in G$ has a fixed reward $u_i(s, a)$ for action $a \in A(s)$ for $s \in S$.
- Each agent i has an expected reward under a policy π in state s:

$$U_i(s, \pi) = \mathbb{E}_{a \sim \pi(s)} \Big[u_i(s, a) \Big]$$

- For each $i \in G$ there is a set of states $N_i \subseteq S$ defined as

$$N_i = \{s \in S : U_i(s, \pi_d) < -\tau\}$$

  for a fixed constant $\tau > 0$.

We will assume the dominant norm stays fixed throughout all of the calculations for simplicity.

The system operates on a discrete time scale where for each $t \in \mathbb{N}$ it enters a state $s(t)$ sampled with $s \sim p(s)$.

3 Defining Resistance Theoretically

Definition 3.1. The grievance of an agent i in a state s is defined as

$$w_i(s) = \max(0, -U_i(s))$$

this measures how dissatisfied an agent i is with the dominant norm in state S.

Remark 3.1. N_i can then be considered the disagreement set of i

Definition 3.2. A subgroup is a pair (H, T) which is a set of agents $H \subseteq G$ and a set of states $T \subseteq S$. This pair is admissible if $T \subseteq \bigcap_{i \in H} N_i$.

Definition 3.3. To measure the strength of a resistant subgroup define

$$\Phi(H, T) = \sum_{i \in H} \sum_{s \in T} p(s) w_i(s)$$

and finding a maximally resistant subgroup requires maximizing $\Phi(H, T)$ for admissible pairs.

We can build a bipartite graph with agents on one side and states on the other with the edge weight between $i \in G$ and $s \in S$ as $W[i, s] = p(s)w_i(s)$. An admissible pair is a biclique, which is a bipartite subgraph in which every agent of H is joined to every state in T. Φ is just the sum of the edge weights. Finding the maximum weight biclique in a bipartite is known to be NP-complete.

We can relax the problem constraints as the hardness of the problem comes from the requirement that $T \subseteq N_i$. We can instead charge each $i \in H$ for the states in T where they benefit by defining

$$V[i, s] = p(s) \max(0, U_i(s, \pi_d))$$

and

$$M_\lambda = W - \lambda V$$

with

$$\Phi_\lambda(H, T) = \sum_{i \in H} \sum_{s \in T} M_\lambda[i, s]$$

where taking $\lambda \to \infty$ recovers the original Φ. In this simplified setup, given fixed T or fixed H the optimal pair is easy to find.

For fixed T the best H is $\{i : r_i > 0\}$ where

$$r_i = \sum_{s \in T} M_\lambda[i, s]$$

and for fixed H the best T is $\{s : c_s > 0\}$ where

$$c_s = \sum_{i \in H} M_\lambda[i, s]$$

4 Subgroup Formation Mechanics

Agents do not have direct access to their rewards but instead keep a running estimate $v_i(s)$ which is initialized at 0. We denote the number of times state s has been visited by $n_i(s)$. At time t. $v_i(s)$ is updated as

$$v_i(s) \to \frac{1}{n_i(s)+1}\Big(u_i(s,a) + n_i(s)v_i(s)\Big)$$

if $s(t) = s$ and otherwise is held constant.

Agents calculate all of the above quantities with $v_i(s)$ in place of $U_i(s)$. When this is the case, we will write $\hat{w}_i(s)$ instead of $w_i(s)$ and etc. We define

$$\hat{N}_i = \{s \in S : n_i(s) \geq n_0, \hat{v}_i(s) < -\tau - \eta\}$$

for a fixed error constant $\eta > 0$ and count n_0.

We use a union bound threshold to implement subgroup formation. The formation process does not happen at every time step t but on a timeline of formation epochs $T(n)$. This is controlled by two parameters: K the number of formation epochs and Δ the spacing between each formation epoch.

Additionally, to prevent premature formation the first formation epoch only occurs after $n_i(s) > n_0$ for all i and s with $p(s) > p_{\min}$ where $p_{\min}$ is a minimum probability for a state to be considered significant.

In other words, formation is only started once all agents have enough context to have accurate estimates.

To determine which agents interact in the formation process we define a similarity metric on the set of agents using Jaccard similarity

$$\text{sim}(i,j) = \frac{\sum_{s\in S} p(s)\min\{\hat{w}_i(s), \hat{w}_j(s)\}}{\sum_{s\in S} p(s)\max\{\hat{w}_i(s), \hat{w}_j(s)\}}$$

then we can build a L nearest neighbours graph on G with the similarity metric measuring closeness.

This graph is recomputed at every formation epoch.

For an agent k define

$$L_k(i) = \sum_{s\in N_k \setminus N_i} p(s)$$

and note the relation

$$p(\bigcap_{i\in H} N_i) \leq p(N_k) - \sum_{i\in H} L_k(i)$$

with the esimated quantity

$$\hat{L}_k(i) = \sum_{s\in \hat{N}_k \setminus \hat{N}_i} p(s)$$

The algorithm for subgroup formation is as follows:

(1) Each candidate agent k collects $N(k)$ the set of its L-nearest neighbours in the Jaccard graph.

(2) Every member of $N(k)$ forwards their L nearest neighbours to k so that its pool of candidates is $N(k) \cup N(N(k))$.

(3) k sorts its pool of candidates by $\hat{L}_k(i)$ and admits a candidate i to its group H_k if

$$\hat{L}_k(i) + \sum_{j \in H_k} \hat{L}_k(j) \leq \epsilon \cdot p(N_k)$$

for some tolerance $\epsilon > 0$. Its final proposed pair is (H_k, N_k) where H_k is the group it formed during this iterative process. Then let $\sigma_k = \Phi_\lambda(H_k, \bigcap_{i \in H_k} \hat{N}_i)$.

(4) Each agent identitifies with the highest scoring proposal including it.

(5) A candidate k commits to H_k if every member of H_k identitifies with its proposal.

(6) Agents that commit to a group are no longer considered in the iteration.

this process occurs at every epoch

5 Theoretical Analysis

I present a theoretical analysis of the abov model to find convergence guarantees under some reasonable assumptions. Most of this section is more mathematically involved, but it only serve to rigorously justify the above definitions. Skip to the discussion at the bottom of this discussion for the relevant conclusions about how the model operates.

First I recap some modelling assumptions stated above: (i) π_d is fixed, (ii) $p(s)$ is known to all agents, (iii) Messages are exact and honest, (iv) rewards $u_i(s, a)$ are bounded by some $B > 0$, (v) At every step the new state is drawn independently from the previous ones, at a visit to s it draws $a \sim \pi_d(s)$ independently and observes $u_i(s, a)$ therefore $\hat{v}_i(s)$ is the mean of $n_i(s)$ i.i.d. draws with mean $U_i(s, \pi_d)$.

A convention is that a leader k is in its own group. Step 3 sorts the pool of candidates by $\hat{L}_k$ and stops at the first candidate that violates the budget, so the admitted set H_k satisfies

$$\sum_{i \in H_k} \hat{L}_k(i) \leq \epsilon p(\hat{N}_k) \tag{1}$$

then k's proposal is $(H_k, \hat{T}_k)$ where $\hat{T}_k = \bigcap_{i \in H_k} \hat{N}_i$ and $\sigma_k = \Phi_\lambda(H_k, \hat{T}_k)$. Ties between proposals are broken by agent id. Leaders are agents with $\hat{N}_k \neq \varnothing$.

Note that for now we assume that each agent acts at every time step, which simplifies the calculations below. However, removing this assumption will not affect the theoretical analysis much qualititatively as only lemma 2 needs to be updated with a larger bound on the minimum count n_0.

Lemma 5.1. *(Estimation). Let* $\theta = \frac{\eta^2}{2B^2}$ *and*

$$n_0 = \lceil \theta^{-1} \log \frac{2|G||S|}{\delta(1 - e^{-\theta})} \rceil$$

then the event

$$E : |\hat{v}_i(s) - U_i(s, \pi_d)| \leq \eta \text{ for all } i, s \text{ with } n_i(s) \geq n_0, \text{ at all times}$$

holds with probability at least $1 - \delta$.

Proof. For fixed (i, s) and n Hoeffding's inequality on the range $[-B, B]$ gives the bound

$$\mathbb{P}(|\bar{u}_{i,n} - U_i(s)| \leq \eta) \leq 2e^{-n\theta}$$

where $\bar{u}_{i,n}$ is the mean of the first draws $u_i(s, \pi_d)$ in the state s which i observes. now we have

$$\begin{aligned}
\mathbb{P}(|\bar{u}_{i,n} - U_i(s)| \leq \eta : n \geq n_0) &\leq \sum_{n=n_0}^{\infty} \mathbb{P}(|\bar{u}_{i,n} - U_i(s)| \geq \eta) \\
&\leq \sum_{n=n_0}^{\infty} 2e^{-n\theta} \\
&= 2e^{-n_0\theta} \sum_{n=0}^{\infty} [e^{-\theta}]^n \\
&= \frac{2e^{-n_0\theta}}{1 - e^{-\theta}}
\end{aligned}$$

and if we plug in the value of n_0 stated above this reduces to

$$\mathbb{P}(|\bar{u}_{i,n} - U_i(s)| \leq \eta : n \geq n_0) \leq \frac{\delta}{|G||S|}$$

then we can find $\mathbb{P}(E)$ as

$$\begin{aligned}
\mathbb{P}(E) &= 1 - \mathbb{P}(E^c) \\
&\geq 1 - \sum_{i,s} \mathbb{P}(|\bar{u}_{i,n} - U_i(s)| \leq \eta : n \geq n_0) \\
&\geq 1 - \sum_{i,s} \frac{\delta}{|G||S|} \\
&= 1 - \delta
\end{aligned}$$

as required. □

Lemma 5.2. *(Visit Counts). For a set of states T with $p_{min}(T) = \min_{s \in T} p(s)$*

$$t \geq \max\{2n_0, 8 \log(|G||S|/\delta)\}/p_{min}(T)$$

then with probability at least $1 - \delta$ every agent has $n_i(s) \geq n_0$. for all $s \in T$ at time t.

Proof. $n_i(s)$ is binomial with mean $tp(s)$. The Chernoff bound then implies

$$\mathbb{P}(n_i(s) \leq tp(s)/2) \leq e^{-tp(s)/8}$$

now we apply a union bound

$$\begin{aligned}
\mathbb{P}(\bigcup_{i,s}\{n_i(s) \le n_0\}) &\le \sum_{i,s}\mathbb{P}(n_i(s) \le n_0) \\
&\text{now since } p(s) \ge p_{\min}\text{:} \\
&\le \sum_{i,s}\mathbb{P}(n_i(s) \le tp(s)/2) \\
&\le \sum_{i,s} e^{-tp(s)/8} \\
&\text{plugging in } tp(s)/8 \ge \log\frac{|G||S|}{\delta}\text{:} \\
&\le \sum_{i,s}\frac{\delta}{|G||S|} \\
&= \delta
\end{aligned}$$

then

$$\mathbb{P}(\bigcup_{i,s}\{n_i(s) > n_0\}) \ge 1-\delta$$

as required. □

Proposition 5.1. *(Soundness) On E the following hold:*

(a) $\hat{N}_i \subseteq N_i$ for every i

(b) For any leader k and any set $H \ni k$

$$p(\bigcap_{i\in H} N_i) \ge p(\bigcap_{i\in H}\hat{N}_i) \ge p(\hat{N}_k) - \sum_{i\in H}\hat{L}_k(i)$$

(c) If $H = H_k$ satisfies (1) then $(H_K, \hat{T}_k)$ is admissible, $p(\hat{T}_k) \ge (1-\epsilon)p(\hat{N}_k)$ and

$$\Phi_\lambda(H, \hat{T}_k) \ge \tau|H|(1-\epsilon)p(\hat{N}_k)$$

Proof. To prove (a) note that if $s \in \hat{N}_i$ then $n_i(s) \ge n_0$, and so on E

$$U_i(s) \le \hat{v}_i(s) + \eta < -\tau$$

to prove (b), since $k \in H$ we have

$$\bigcap_{i\in H}\hat{N}_i = \hat{N}_k \setminus \bigcup_{i\in H}(\hat{N}_k \setminus \hat{N}_i)$$

□